access is controlled by traditional methods such as those listed in the preceding paragraph. But blockchain is a decentralized system. In such a system, the participants typically are distributed, hold their assets themselves, can join and leave as they wish (within the rules coded in the smart contracts), have a self-managed identity, are loosely organized, and depend on the blockchain for the trust layer. Under these conditions, establishing identity and ensuring privacy and security are indeed challenging--but in this chapter, you'll learn methods for application of cryptography and hashing to address these issues in decentralized systems.

Remember the quad charts from chapter 3? In the diagram (repeated here in figure 5.1), the components of trust in the chart on the left--verification, validation, and recording with consensus--is addressed by smart contract modifiers and transaction recording on the blockchain (chapter 3). The issues in the quad chart on the right--identity, security, privacy, and confidentiality--are grouped (in the decentralized context) as contributors to the integrity of the system. Our focus in this chapter will be on addressing identity (2a), security (2b), and privacy (2c) concerns.

